Zuclopenthixol decanoate and haloperidol decanoate in chronic schizophrenia: a double-blind multicentre study

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Zuclopenthixol decanoate (ZPT-D) and haloperidol decanoate (HAL-D) were compared in the maintenance treatment of chronic schizophrenic patients. All patients were treated with either ZPT-D or HAL-D for at least 3 months and were then randomly allocated to treatment with either ZPT-D (100–600 mg every 4 weeks) or HAL-D (38–200 mg every 4 weeks) for 9 months. Sixty-four patients entered the study. Three patients were only assessed at baseline. Four patients in the HAL-D group were withdrawn because of deterioration. One patient in the ZPT-D group committed suicide. Fifty-six patients completed the trial. The assessment programme comprised Clinical Global Impressions by psychiatrists and nurses, the Brief Psychiatric Rating Scale (BPRS), Montgomery-Åsberg Depression Rating Scale (MADRS), UKU Side Effect Scale and Simpson-Angus Scale. Ratings were made at baseline and after 12, 24 and 36 weeks of treatment. The severity of illness scores remained almost constant. The only between-group difference was recorded at month 6 in favour of ZPT-D. The BPRS total scores were reduced significantly in both groups with no between-group differences. The depression scores on the MADRS were very low. Both treatments caused few and mild side effects according to the UKU Side Effect Scale and the Simpson-Angus scale, and there were no significant differences between the groups. Both ZPT-D and HAL-D seem to be effective in the maintenance treatment of chronic schizophrenic patients and cause few side effects. The injections of ZPT-D and HAL-D can be given at 4-week intervals.

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Long-acting depot neuroleptics have definite advantages over oral formulations in the maintenance treatment of chronic schizophrenic patients. Depot injections overcome the problems of first-pass metabolism by the liver and give a more predictable and constant serum level. However, this is achieved at expense of a reduced flexibility of prescription. The most important clinical advantage of depot administration is the improved compliance. The optimal compliance probably accounts for the much lower relapse rate in patients receiving depot injections than in patients receiving oral medication (1, 2).

Haloperidol and clopenthixol have been used since about 1960 in the treatment of psychoses (3, 4). Later clopenthixol was replaced by zuclopenthixol, which is the neuroleptically active isomer of clopenthixol (5). The short-acting oral formulations of zuclopenthixol and haloperidol have been compared both in the treatment of acute psychoses (6) and in the maintenance treatment of chronic schizophrenia (7). The 2 neuroleptics were found to be equally effective, but extrapyramidal side effects were seen more frequently in patients treated with haloperidol. Recently a new short-acting formulation for intramuscular use, zuclopenthixol acetate dissolved in thin vegetable oil, was compared with haloperidol in the initial treatment of acutely psychotic patients in need of parenteral neuroleptic therapy (8). The 2 treatments were equally effective in patients with an acute psychosis or an exacerbation of a chronic psychosis, but zuclopenthixol acetate caused a more rapid remission of symptoms in manic patients. Also in this study, haloperidol induced a larger number extrapyramidal symptoms.

Depot formulations of both zuclopenthixol and haloperidol have been developed and both have been shown to be effective in managing chronic schizophrenia (9–12).

The aim of this study was to evaluate and compare the wanted and unwanted effects of zuclopenthixol decanoate and haloperidol decanoate in the maintenance of chronic schizophrenic patients.

Material and methods

Design

The study was conducted as a double-blind, multicentre study including outpatient clinics in Finland and Sweden. The protocol was approved by the local ethics committees. Both depot antipsychotics were given at 4-week intervals and the 9-month treatment period was divided into 2 parts. Period A was a 3-month dose-adjustment period and supplementary injections could be given at weeks 2 and 6, if clinically indicated. During the last 6 months, period B, the patients continued the double-blind treatment and the dosage was kept as constant as possible.

Patients

Outpatients of either sex aged between 25 and 60 years with a diagnosis of chronic schizophrenia according to DSM-III were considered for inclusion. All patients had to be stabilized on either zuclopenthixol decanoate or haloperidol decanoate for a minimum of 3 months, and the total score on the 18-item Brief Psychiatric Rating Scale (BPRS) had to be between 5 and 26. Excluded were patients with a total duration of illness less than 2 years or a duration of hospitalization of more than 10 years. Drug or alcohol abuse, pregnancy, known organic brain disorder or uncompensated somatic illness were also exclusion criteria. Also, patients receiving doses higher than 600 mg zuclopenthixol decanoate or 150 mg haloperidol decanoate per injection were excluded. Patients could be withdrawn from the study in case of deterioration of their psychotic state or in case of troublesome side effect. Patients who wanted to leave the study were free to do so.

Dosage

The commercially available formulations of zuclopenthixol decanoate (Cisordinol® Depot) and haloperidol decanoate (Haldol®, Serenase®, Haloperin® Depot) were used. Two hundred mg zuclopenthixol decanoate (1 ml) was considered to be equivalent to 50 mg haloperidol decanoate (1 ml). The injections were given at 4-week intervals and the doses were individually tailored according to the patient's need. The first dose was to be equivalent with the last dose in the pre-trial period.

As the 2 depot preparations are not quite identical in appearance, the so-called triple-blind technique was used to keep blindness. The injections were given by a nurse who had access to the code, but was not involved in the assessment of the patients. Concomitant medication should be restricted to a minimum. In period A levomepromazine in daily doses of 100 mg or lower might be given, but only for up to 14 d in each 4-week period.Patients with occasional restlessness could be given lorazepam or oxazepam, and orphenadrine or biperiden could be prescribed in case of extrapyramidal side effects. Sleep disturbances could be treated with chloral hydrate or oxazepam, and amitriptyline could be given to patients with depression. Patients receiving lithium could continue this treatment unchanged throughout the study. Drugs for the treatment of somatic diseases could also be prescribed.

Assessment

The patients were assessed at baseline and during the study according to the assessment programme shown in Fig. 1. Severity of illness was assessed both by psychiatrists and by members of the nursing staff according to the Clinical Global Impression (CGI) scale (13). Furthermore, all patients were rated on the BPRS (14) and on the Montgomery-Åsberg Depression Rating Scale (MADRS) (15). Unwanted effects were assessed by means of the UKU Side Effect Scale (16) and the Simpson-Angus Scale (17). To improve interrater reliability, joint ratings were arranged before and during the trial.

Laboratory investigations comprising liver and kidney function tests and red blood cell and white blood cell counts were made before and after test treatment. Blood samples for zuclopenthixol and haloperidol assays were taken at week 36 and zuclopenthixol and haloperidol were measured as described by Aaes-Jørgensen (18).

Assessment programme

Rating scales	Baseline	Week 12	Week 24	Week 36
	Period A		Period B	
CGI-Psychiatrists	x	x	x	x
CGI-Nurses	x	x	x	x
BPRS	x	x	x	x
MADRS	x	x	x	x
UKU	x	x	x	x
Simpson & Angus	x	x	x	x

Fig. 1. Survey of the assessments throughout the trial period

Statistical analysis

To ensure that the results could be analysed as planned in the protocol, an analysis of covariance (repeated measures) using baseline scores and pre-trial antipsychotic treatment as covariates was made. Paired *t*-test was used to estimate within-group differences and between-group differences were estimated by unpaired *t*-test. The SAS computer package was used (19).

Results

A total of 64 patients (35 from Finland and 29 from Sweden) entered the study. The average age of the Finnish and Swedish patients was very similar; 40 (range 21–61) and 38 years (range 27–55), respectively. The Finnish and Swedish patients also had similar total duration of illness and baseline severity of illness scores.

Two patients allocated to zuclopenthixol decanoate were withdrawn for administrative reasons unrelated to test treatment and 1 patient allocated to haloperidol decanoate was withdrawn because of side effects. These 3 patients were only assessed at baseline. This leaves 61 patients for the statistical analyses. Of these, 34 were treated with zuclopenthixol decanoate and 27 with haloperidol decanoate. One patient in the zuclopenthixol group, a 28-year-old women, committed suicide after 12 weeks of treatment. Of the 27 haloperidol patients, 4 withdrew prematurely because of deterioration.

The distribution of the patients according to sex and age is shown in Table 1. The distributions according to age were very similar in the treatment groups, but more women were treated with zuclopenthixol decanoate. Concerning the total duration of illness, duration of outpatient treatment and duration of previous hospitalization, there were no relevant differences between the treatments. Most patients had been ill for more than 10 years and had been treated as outpatients for 2–10 years.

Table 1. Patients distributed by age and sex (*n*)

	Drug						
	Zuclopenthixol			Haloperidol			
	Sex			Sex			
Age (years)	M	F	Total	M	F	Total	Total
20–29	3	4	7	2	3	5	12
30–39	7	5	12	7	4	11	23
40–49	2	9	11	5	4	9	20
≥50	1	3	4	1	1	2	6
Total	13	21	34	15	12	27	61
Average age	37	40	39	38	38	38	38

Medication

The average dose per injection of zuclopenthixol decanoate at week 36 was 284 mg (range 100–600) and that of haloperidol decanoate 92 mg (range 38–200). The intervals between the injections were on average about 4 weeks.

Four patients in the zuclopenthixol group and 2 patients in the haloperidol group were given amitryptyline during some of the trial period. Antiparkinsonian drugs were given to 5 patients throughout the study period (3/34 zuclopenthixol, 2/27 haloperidol) and to 1 patient (zuclopenthixol) for only 1 month. About one third of the patients in both treatment groups received oxazepam at some time.

Therapeutic effect

The results of the global assessment of severity of illness by the psychiatrists are shown in Fig. 2. The average baseline score was 3.9 in both groups. A slight reduction in scores was obtained in the zuclopenthixol decanoate group, whereas the scores of the haloperidol decanoate patients remained almost unchanged during the 36-week treatment period. At the 24-week assessment, there was a difference between the groups in favour of zuclopenthixol decanoate ($P = 0.02$).

The results of the nurses' assessment of severity of illness showed that the average baseline score in the zuclopenthixol decanoate group was 4.4, and the scores at all ratings during treatment were slightly lower than the baseline score ($P < 0.01$). The average baseline score in the haloperidol decanoate group was 4.2 and the average scores in this group remained almost constant throughout the trial period. The differences in average scores of the groups were not statistically significant.

The analysis of covariance (repeated measures) of the BPRS total scores showed that baseline scores significantly influenced the later scores ($P < 0.001$), whereas a significant influence by pre-trial treatment could not be demonstrated. Fig. 3 shows the average total scores in the 2 treatment groups. The average baseline scores in the groups were rather similar and the scores were reduced during test treatment. In the zuclopenthixol decanoate group, the reduction was statistically significant at all ratings, whereas this was the case only after 36 weeks of treatment with haloperidol decanoate. There were no significant between-group differences.

A change in BPRS total score of 3 or more is

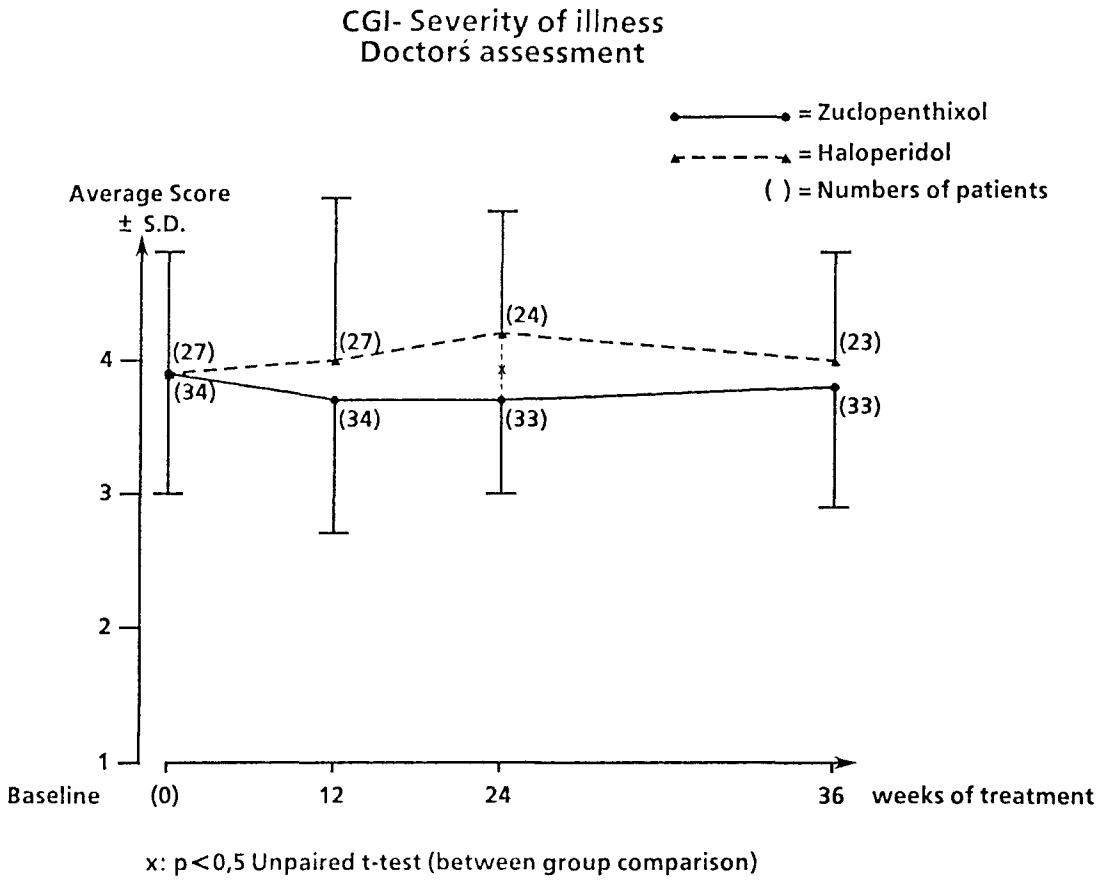


Fig. 2. Average severity of illness scores according to the CGI. Assessments by psychiatrists. The number in parentheses indicates number of patients. Vertical bars indicate standard deviations.

considered to be clinically relevant. By comparing baseline scores and final scores, 4 patients (3 males and 1 female) deteriorated, 11 patients (5 males and 6 females) remained unchanged and 18 patients (5 males and 13 females) improved during treatment with zuclopenthixol decanoate. For haloperidol decanoate, 3 men deteriorated, 14 patients (6 males and 8 females) were unchanged and 10 patients (7 males and 3 females) improved.

A symptom group analysis as described by Guy (13) showed that the factor scores generally were similar in the treatment groups. The "anxiety-depression" score of the zuclopenthixol decanoate group at baseline was significantly higher than that of the haloperidol decanoate group. However, later in the experimental period there were no differences between the groups. The symptom group analysis did not disclose any qualitative differences between the treatments.

The average MADRS total scores at baseline were very low: 1.1 (SD: 1.5) in the zuclopenthixol decanoate group and 1.5 (SD: 2.7) in the haloperidol decanoate group. The changes in scores did not reach the level of statistical significance in any of the groups and there were no between-group differences.

Side effects

The results of the global assessment of side effects showed that the majority of patients had no or only mild side effects without influence on the patients' functioning during the entire trial period. At baseline, i.e. at the end of pre-trial treatment, 32 of the patients allocated to treatment with zuclopenthixol decanoate were assessed. Of these, 25 (78%) had no or only mild side effects (scores 1 or 2). After 36 weeks on zuclopenthixol decanoate, 27 of 32 patients (84%) had no or only mild side effects. Of the patients allocated to haloperidol decanoate, 24 of 25 patients (96%) had scores 1 or 2 at baseline. After 36 weeks of treatment the corresponding figures were 24 of 26 patients (92%).

Single side effects were evaluated by using the UKU Side Effect Scale and the results showed that both treatments caused few and in most cases mild side effects. The most commonly recorded mental side effects were concentration difficulties and sleepiness or sedation. The number of central nervous system (CNS) side effects was low in both groups. At the week-24 assessment, 2 cases of mild dystonia were recorded in the zuclopenthixol decanoate

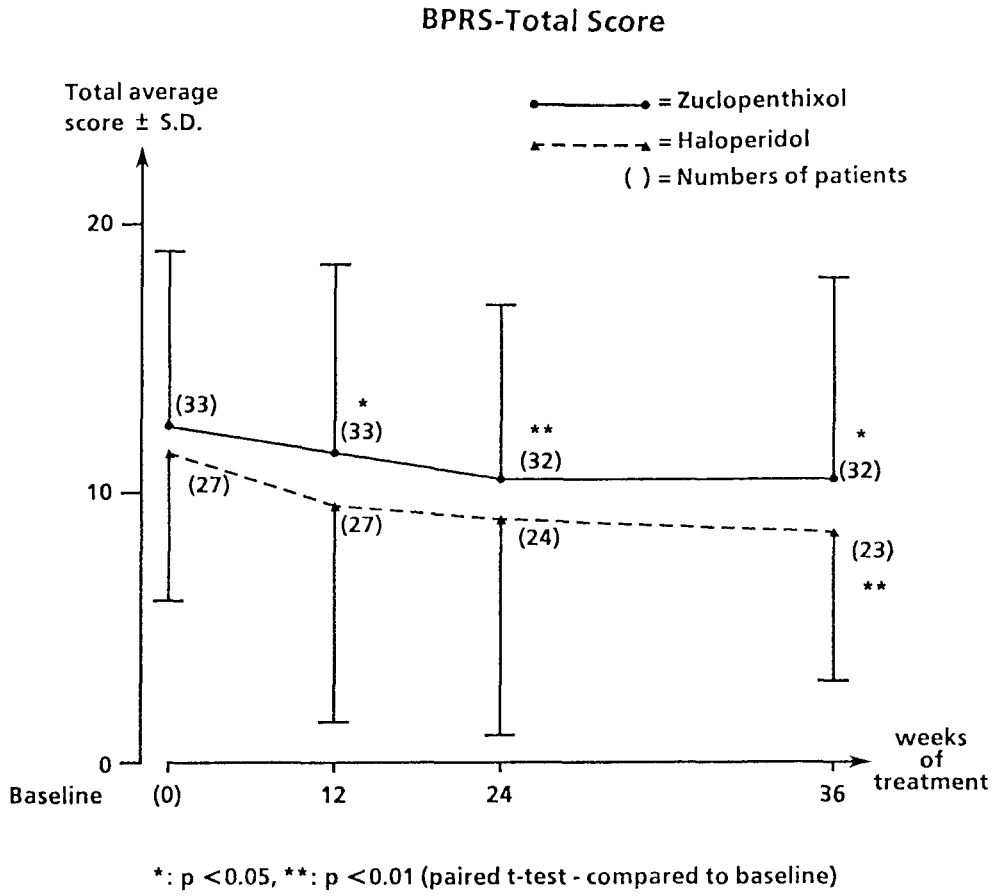


Fig. 3. Average BPRS scores. The number in parentheses indicates number of patients. Vertical bars indicate standard deviations.

group and 2 cases of mild and one case of moderate dystonia were observed in the haloperidol decanoate group. The most commonly recorded autonomic symptoms were palpitations, orthostatic dizziness, and accommodation disturbances. Table 2 shows the number of mental, CNS, and autonomic symptoms observed in the treatment groups. The number of mental symptoms tended to decrease during test treatment and the number of CNS symptoms was low throughout the period. The number of autonomic symptoms observed in the patients treated with zuclopenthixol decanoate decreased clearly over the 36 weeks, whereas the opposite was the case in the patients treated with haloperidol decanoate.

The extrapyramidal symptoms were also assessed by using the Simpson-Angus Scale. The total scores were low, about 1 in each of the treatment groups with no between-group differences. Pre-trial treatment had no influence on the total scores (analysis of covariance).

In the Swedish patients the possible appearance of tardive dyskinesia was studied by using the Abnormal Involuntary Movement Scale (13). Tardive dyskinesia was observed in 2 patients during treat-

Table 2. Number of probably drug-induced side effects – UKU Side Effect Scale

	Baseline		Week 12		Week 24		Week 36	
Type of symptom	ZPT-D n=34	HAL-D n=27	ZPT-D n=33	HAL-D n=24	ZPT-D n=33	HAL-D n=24	ZPT-D n=33	HAL-D n=23
Mental	12	11	14	9	8	10	5	7
CNS	8	1	2	0	5	4	4	1
Autonomic	22	9	13	16	13	22	8	29

ZPT-D: zuclopenthixol decanoate; HAL-D: haloperidol decanoate.

ment with zuclopenthixol decanoate and in 4 patients during treatment with haloperidol decanoate. Five of the patients had tardive dyskinesia at baseline and 1 haloperidol patient developed a mild dyskinesia after 24 weeks.

Both before and after treatment, some laboratory values were outside the normal range of the laboratories of the participating hospitals. A few cases of elevated serum ALAT and serum alkaline phosphatase values were seen, but in no case were the changes considered to be clinically relevant.

Serum levels of zuclopenthixol at week 36 were measured in blood samples from 13 patients and haloperidol concentrations in samples from 22 patients. The mean zuclopenthixol concentration was 7.4 nmol/l (range 1.8–15.0). There was a correlation between dose and serum concentration ($r = 0.69$, $P < 0.01$). The mean serum concentration of haloperidol was 6.0 nmol/l with the range 0.7–16.5. Also in this group, dose and serum concentration were correlated ($r = 0.80$, $P < 0.001$).

Discussion

The results of this double-blind comparative study seem to confirm that both zuclopenthixol decanoate and haloperidol decanoate are effective in the treatment of chronic schizophrenia. The results also indicate that the injections in many cases can be given at 4-week intervals. Only 4 patients, all receiving haloperidol decanoate, deteriorated during this treatment. It is not unlikely that these patients would have done better if higher doses had been given or if the injections had been given at shorter intervals.

The BPRS total scores were slightly reduced in both treatment groups. This reduction of symptom scores may be caused by a beneficial effect of the depot neuroleptics. However, it is likely that the closer observation of the patients during the trial situation may have contributed as well. The reduction in BPRS scores was reflected in a slight reduction in the global severity of illness scores in patients receiving zuclopenthixol decanoate, but not in patients treated with haloperidol decanoate. This indicates that the members of the staff – psychiatrists as well as nurses – have found the improvement slightly greater in the patients treated with zuclopenthixol decanoate.

There were somewhat more females (13 of 18) among the patients who improved during treatment with zuclopenthixol decanoate than among the patients who improved during treatment with haloperidol decanoate (3 of 10). Generally, women are known to respond better to neuroleptic treatment than men because of better compliance. As all patients in this study received depot injections, compliance was not a problem and the apparently higher number of women responders in the zuclopenthixol decanoate group is probably an incidental finding.

The importance of depression in the life of a schizophrenic patient has only been realized during the last decade, and reviews of the literature have concluded that depression in schizophrenia is frequent (16, 20, 21). The very low MADRS scores in the patients of both treatment groups show that most patients in this study were not depressed. However, one patient committed suicide after 13 weeks. At the assessment at week 12, the patient had a BPRS total score of 10 and an MADRS total score of 14. The corresponding baseline scores were 9 and 10, respectively. This shows that the patient's condition had remained almost unchanged and that the patient was only mildly depressed (22). Although the exact relationship between suicide and clinical depression is uncertain, it is known that the risk of suicide in schizophrenia is high (23, 24).

Short-acting formulations of zuclopenthixol are known to induce a somewhat stronger sedation and fewer extrapyramidal symptoms than the short-acting formulations of haloperidol (8). A similar difference in clinical profile of the depot formulations was not found. It is likely that tolerance has developed to the sedative action of zuclopenthixol and it is also possible that a certain adaptation to the extrapyramidal symptoms expected to be induced by haloperidol has developed over the long period of treatment. The low frequency of extrapyramidal side effects cannot be explained by frequent prescription of antiparkinsonian agents, as only few patients received such drugs. It is also interesting that tardive dyskinesia was only observed in very few of the patients in this study, and in only 1 patient did the dyskinesia develop during test treatment.

In the zuclopenthixol group the number of autonomic side effects decreased during the 36-week test treatment. This is an expected finding, but it is surprising that the number of autonomic symptoms increased in the haloperidol group. It is unlikely that these side effects have been induced by concomitant medication such as antiparkinsonian drugs, since only 2 patients in the haloperidol group received antiparkinsonian agents, and we have no explanation for this finding.

Although a detailed analysis of the possible correlation between serum zuclopenthixol and haloperidol levels and clinical effects has not been included in this article (to be published elsewhere), there was no obvious correlation between the serum levels and wanted and unwanted effects. Because of the very low BPRS scores, it probably is difficult to demonstrate a possible correlation between the therapeutic effect and serum concentrations. A correlation between serum concentrations and side effects might be easier to demonstrate if side effect assessments

had been made and blood samples taken at the time of peak concentration, about 1 week after the injection. However, this was not included in the design of this study.

According to the study design, the dosage adjustment period was 3 months, which is a short period with long-acting neuroleptics. This means that it is not certain that all patients have been given the lowest effective dose.

The commercially available formulation of zuclopenthixol decanoate is a solution of zuclopenthixol decanoate in thin vegetable oil (Viscoleo®) whereas haloperidol is dissolved in sesame oil. One might speculate that the different vehicles might result in different pharmacokinetic properties (25). However, pharmacokinetic studies have shown that the release half-lives of zuclopenthixol decanoate and haloperidol decanoate are very similar, 19 and 21 d respectively (26). It is therefore not surprising that both depot drugs can be given at 4-week intervals. This is in good agreement with the experience gained in our daily clinical work. About 80% of our chronic schizophrenic patients are given their depot injections once a month (Wistedt et al., personal communication).

The effect of the test treatment in this study did not depend on the treatment the patients received before entry into the study. This is in accordance with the fact that patients can usually be switched from one neuroleptic drug to another, provided that the doses are equivalent and the clinical profiles are similar.

In conclusion, the results seem to confirm that zuclopenthixol decanoate and haloperidol decanoate are effective in maintaining improvement in chronic schizophrenic patients. The two depot antipsychotics are about equally effective and both can be given at 4-week intervals. Side effects are generally few and mild.

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References

1. GOTTFRIES CG, GREEN L. Flupenthixol decanoate in treatment of outpatients. Acta Psychiatr Scand 1974: suppl 255: 15–23.
2. JOHNSON DAW. The expectation of outcome for maintenance therapy in chronic schizophrenic patients. Br J Psychiatry 1976: 128: 246–250.
3. JANSSEN PAJ, VAN BEVER WFM. Butyrophenones and diphenylbutylamines. In: USDIN E, FORREST IS, ed. Psychotherapeutic drugs. Part 2, Applications (Psychopharmacology series, 2). New York: Marcel Dekker, 1977: 869–921.
4. PETERSEN PV, MØLLER NIELSEN I, PEDERSEN V, JØRGENSEN A, LASSEN N. Thioxanthenes. In: USDIN E, FORREST IS, ed. Psychotherapeutic drugs. Part 2, Applications (Psychopharmacology series, 2). New York: Marcel Dekker, 1977: 827–867.
5. GRAVEM A, ELGEN K. Cis(Z)-clopenthixol. The neuroleptically active isomer of clopenthixol. Acta Psychiatr Scand 1981: 64: suppl 294.
6. KUNY S, WOGGON B. Cis(Z)-clopenthixol in the treatment of acute schizophrenia. In: HALL P, ed. Modern trends in the chemotherapy of schizophrenia. VII World Congress of Psychiatry, Vienna. Proceedings of a special symposium held 13 July 1983. Copenhagen: H. Lundbeck A/S, 1983: 15–24.
7. HEIKKILÄ L, LAITINEN J, VARTIAINEN H. Cis(Z)-clopenthixol and haloperidol in chronic schizophrenic patients – a double-blind clinical multicentre investigation. Acta Psychiatr Scand 1981: 64 (suppl 294): 30–38.
8. BOBON D, DE BLEEKER E. Zuclopenthixol acetate and haloperidol in acute psychotic patients. A randomized multicentre study. Presented at the ECNP Congress in Gothenburg, Sweden May 24–26, 1989.
9. AHLFORS UG, DENCKER SJ, GRAVEM A, REMVIG J. Clopenthixol decanoate and perphenazine enanthate in schizophrenic patients. A double-blind Nordic multicentre trial. Acta Psychiatr Scand 1980: 61 (suppl 279): 77–91.
10. DENCKER SJ, LEPP M, MALM U. Clopenthixol and flupenthixol depot preparations in outpatient schizophrenics. I. A one year double-blind study of clopenthixol decanoate and flupenthixol palmitate. Acta Psychiatr Scand 1980: 61 (suppl 279): 10–28.
11. CHOUINARD G, ANNABLE L, CAMPBELL W, BOISVERT D, BRADWEJN J. A double-blind controlled clinical trial of haloperidol decanoate and fluphenazine decanoate in the maintenance treatment of schizophrenia. Psychopharmacol Bull 1984: 20: 108–109.
12. WISTEDT B. A comparative trial of haloperidol decanoate and fluphenazine decanoate in chronic schizophrenic patients. Int Clin Psychopharmacol 1986: 1: suppl 1: 15–23.
13. GUY W. Early clinical drug evaluation (ECDEU) assessment manual for psychopharmacology. National Institute of Mental Health Publication No. 76–338. Washington, DC: US Government Printing Office, 1976.
14. BECH P, KASTRUP M, RAFAELSEN OJ. Mini-compendium of rating scales for states of anxiety, depression, mania, schizophrenia with corresponding DSM-III syndromes. Acta Psychiatr Scand 1986: 73: suppl 326.
15. MONTGOMERY SA, ASBERG M. A new depression scale designed to be sensitive to change. Br J Psychiatry 1979: 134: 382–389.
16. LINGJÆRDE O, AHLFORS UG, BECH P, DENCKER SJ, ELGEN K. The UKU side effect rating scale. A new comprehensive rating scale for psychotropic drugs and a cross-sectional study of side effects in neuroleptic-treated patients. Acta Psychiatr Scand 1987: 76: suppl 334.
17. SIMPSON GM, ANGUS JWS. A rating scale for extrapyramidal side effects. Acta Psychiatr Scand 1970: suppl 212: 11–19.
18. AAES-JØRGENSEN T. Specific high-performance liquid chromatographic method for estimation of the cis(Z) – and trans(E) – isomers of clopenthixol and a N-dealkyl metabolite. J Chromatogr 1980: 183: 239–245.
19. SAS user's guide: statistics. Cary, NC: SAS Institute, 1985.
20. KNIGHTS A, HIRSCH SR. "Revealed" depression and drug treatment for schizophrenia. Arch Gen Psychiatry 1981: 38: 806–811.
21. WISTEDT B, PALMSTIERNA T. Depressive symptoms in chronic schizophrenic patients after withdrawal of long-term neuroleptics. J Clin Psychiatry 1983: 44: 369–371.

22. SNAITH RP, HARROP FM, NEWBY DA, TEALE C. Grade scores of the Montgomery-Åsberg Depression and the Clinical Anxiety Scales. Br J Psychiatry 1986: 148: 599–601.
23. ROY A. Suicide in chronic schizophrenia. Br J Psychiatry 1982: 141: 171–177.
24. ALLEBECK P, WISTEDT B. Mortality in schizophrenia. Arch Gen Psychiatry 1986: 43: 650–653.
25. WISTEDT B, JØRGENSEN A, WILES D. A depot neuroleptic withdrawal study. Plasma concentrations of fluphenazine and flupenthixol and relapse frequency. Psychopharmacology 1982: 78: 301–304.
26. JANN MW, ERESHEFSKY L, SAKLAD SR. Clinical pharmacokinetics of the depot antipsychotics. Clin Pharmacokinet 1985: 10: 315–333.